virtue of changes in their mutual gravitational attraction, the influence will not be instantaneously communicated. Rather, as an object changes its position or even blows apart, it causes a change in the distortion of the spacetime fabric that spreads outward at light speed, precisely in keeping with the cosmic speed limit of special relativity. Thus, we on earth would visually learn of the sun's destruction at the same moment that we would feel the gravitational consequences—about eight minutes after it explodes. Einstein's formulation thereby resolves the conflict; gravitational disturbances keep pace with, but do not outrun, photons.

The Warping of Time, Revisited

Illustrations such as those of Figures 3.2, 3.4, and 3.6 capture the essence of what "warped space" means. A warp distorts the shape of space. Physicists have invented analogous images to try to convey the meaning of "warped time," but they are significantly more difficult to decipher, so we will not introduce them here. Instead, let's follow up the example of Slim and Jim on the Tornado ride, and try to get a sense of the experience of gravitationally induced warped time.

To do so, we revisit George and Gracie, no longer in the deep darkness of empty space, but floating near the outskirts of the solar system. They are still each wearing large digital clocks on their space suits that are initially synchronized. To keep things simple, we ignore the effects of the planets and consider only the gravitational field of the sun. Let's further imagine that a spaceship hovering near George and Gracie has reeled out a long cable extending all the way down to the vicinity of the sun's surface. George uses this cable to slowly lower himself toward the sun. As he does so, he periodically stops so that he and Gracie can compare the rate at which time is elapsing on their clocks. The warping of time predicted by Einstein's general relativity implies that George's clock will run slower and slower compared with Gracie's as the gravitational field he experiences gets stronger and stronger. That is, the closer he gets to the sun the slower his clock will run. It is in this sense that gravity distorts time as well as space.

You should note that unlike the case in Chapter 2 in which George and Gracie were in empty space moving relative to each other with constant velocity, in the present setting there is no symmetry between them. George, unlike Gracie, feels the force of gravity getting stronger